

R Package Validation Report — Tier 1 Screening

jsonlite 2.0.0

Package jsonlite
Version 2.0.0 [EV-002]
Origin cran [EV-009]
Tarball sha256 75eb910c82b350ec33f094779da0f87bff154c232e4ae39c9896a9b89f3ac82d [EV-008]
Intended-use tier gxp-support
Report generated (UTC) 2026-09-05T19:53:19Z

Decision

GO-WITH-CONDITIONS — overall risk score 0.2829 [EV-022]
 0 = low risk, 1 = high; config v1.1.0

Rules evaluated (fired or not, with inputs and thresholds):

Rule	Input	Condition	Fired
overall_score_threshold	0.2829	score $\geq$ 0.2000 / 0.4000	FIRED
news_missing	1	$\geq$ 0.5000	FIRED
news_not_measurable	1	is_na	no
low_downloads	1620938	$\geq$ 1000	no
bugs_mostly_open	0.7667	$\geq$ 0.5000	FIRED
no_source_control	jeroen/jsonlite	is_na	no
exports_undocumented	1	$\geq$ 0.5000	FIRED
license_risk	(no input)	$\geq$ 0.0000	no
heavy_dependencies	0.8176	$\geq$ 0.5000	FIRED

Conditions

- NEWS not current — pin the assessed version and re-check on update.
- Majority of recent issues unresolved — review the bug tracker before use.
- Exported functions lack documentation — require documentation review for functions in scope.
- Large dependency footprint — each dependency needs its own screening decision.

Per-Metric Breakdown ({riskmetric})

Metric	Risk (0–1)	Evidence
bugs_status	0.7667	EV-023
dependencies	0.8176	EV-024
downloads_1yr	0.9920	EV-025
export_help	1	EV-026
exported_namespace	0.5128	EV-027
has_bug_reports_url	1	EV-028
has_examples	1	EV-029
has_maintainer	1	EV-030

Metric	Risk (0–1)	Evidence
has_news	1	EV-031
has_source_control	0	EV-032
has_vignettes	1	EV-033
has_website	1	EV-034
news_current	1	EV-035
reverse_dependencies	1	EV-036
size_codebase	0.1020	EV-037

Scoring Strategy & Category Subscores

Category	Weight	Subscore (risk)	(0–1)	Member metrics
code	0.5000	0.4775		dependencies: 0.8176 [EV-024], exported_namespace: 0.5128 [EV-027], remote_checks: NA, size_codebase: 0.1020 [EV-037]
documentation	0.1500	1.0000		export_help: 1 [EV-026], has_examples: 1 [EV-029], has_news: 1 [EV-031], has_vignettes: 1 [EV-033], has_website: 1 [EV-034], license: NA, news_current: 1 [EV-035]
maintenance	0.2000	0.6917		bugs_status: 0.7667 [EV-023], has_bug_reports_url: 1 [EV-028], has_maintainer: 1 [EV-030], has_source_control: 0 [EV-032]
popularity	0.1500	0.9960		downloads_1yr: 0.9920 [EV-025], reverse_dependencies: 1 [EV-036]

Rubric: each category subscore is the equal-weighted mean of its member metrics' non-null riskmetric scores (0 = low risk, 1 = high risk); weights disclose the relative importance of each category. Category subscores are display-only — the overall score and the decision thresholds are unchanged.

Classification & Sign-off

Not classified — no intended use or justification was recorded for this assessment.

Risk classification: PENDING HUMAN REVIEW — this tool does not classify.

Sign-off: PENDING HUMAN REVIEW — machine-generated screening output; human QA sign-off pending. This tool never signs.

Evidence Register

ID	Metric	Value	Mode
EV-001	description.package	jsonlite	local
EV-002	description.version	2.0.0	local
EV-003	description.title	A Simple and Robust JSON Parser and Generator for R	local
EV-004	description.license	MIT + file LICENSE	local
EV-005	description.maintainer	Jeroen Ooms jeroenooms@gmail.com	local
EV-006	description.url	https://jeroen.r-universe.dev/jsonlite	local
EV-007	description.bug_reports	https://arxiv.org/...	local
EV-008	tarball.sha256	https://github.com/jeroen/jsonlite/issues 75eb910c82b350ec33f094779da0f87bff154c232e4e39c9896a9b89...	local

ID	Metric	Value	Mode
EV-009	ingest.origin	cran	local
EV-010	ingest.source_url	https://cran.r-project.org/src/contrib/jsonlite.2.0.0.tar.gz	local
EV-011	cran.version	2.0.0	network
EV-012	cran.title	A Simple and Robust JSON Parser and Generator for R	network
EV-013	cran.maintainer	Jeroen Ooms jeroenooms@gmail.com	network
EV-014	cran.license	MIT + file LICENSE	network
EV-015	cran.url	https://jeroen.r-universe.dev/jsonlite https://arxiv.org/...	network
EV-016	downloads.last_month_total	1620938	network
EV-017	github.repo	jeroen/jsonlite	network
EV-018	github.stars	403	network
EV-019	github.forks	45	network
EV-020	github.open_issues	122	network
EV-021	github.last_push	2026-09-03T13:26:51Z	network
EV-022	riskmetric.overall_score	0.2829	riskmetric
EV-023	riskmetric.score.bugs_status	0.7667	riskmetric
EV-024	riskmetric.score.dependencies	0.8176	riskmetric
EV-025	riskmetric.score.downloads_1yr	0.9920	riskmetric
EV-026	riskmetric.score.export_help	1	riskmetric
EV-027	riskmetric.score.exported_namespace	0.5128	riskmetric
EV-028	riskmetric.score.has_bug_reports_url	1	riskmetric
EV-029	riskmetric.score.has_examples	1	riskmetric
EV-030	riskmetric.score.has_maintainer	1	riskmetric
EV-031	riskmetric.score.has_news	1	riskmetric
EV-032	riskmetric.score.has_source_control	0	riskmetric
EV-033	riskmetric.score.has_vignettes	1	riskmetric
EV-034	riskmetric.score.has_website	1	riskmetric
EV-035	riskmetric.score.news_current	1	riskmetric
EV-036	riskmetric.score.reverse_dependencies	1	riskmetric
EV-037	riskmetric.score.size_codebase	0.1020	riskmetric
EV-038	riskmetric.raw.downloads_1yr	18506916	riskmetric
EV-039	riskmetric.raw.has_bug_reports_url	1	riskmetric
EV-040	riskmetric.raw.has_maintainer	Jeroen Ooms jeroenooms@gmail.com	riskmetric
EV-041	riskmetric.raw.has_news	1	riskmetric
EV-042	riskmetric.raw.has_vignettes	18	riskmetric
EV-043	riskmetric.raw.has_website	https://jeroen.r-universe.dev/jsonlite https://arxiv.org/...	riskmetric
EV-044	riskmetric.raw.license	MIT + file LICENSE	riskmetric
EV-045	riskmetric.raw.news_current	TRUE	riskmetric
EV-046	riskmetric.raw.size_codebase	1320	riskmetric
EV-047	riskmetric.excluded_metrics	assess_covr_coverage, sess_r_cmd_check	as-riskmetric

Tier-2 Sandbox Validation

Deep validation in a throwaway R sandbox (package code executed in isolation): R CMD check, the package's own test suite and test coverage, pinned to the recorded CRAN snapshot.

Attention: the sandboxed validation did not come back clean (status check-errors); review the counts below.

Sandbox status	check-errors
Tests (passed/total)	63 / 68 (failed 4, skipped 1)
R CMD check	errors (errors 1, warnings 1, notes 1)
Test coverage	65.2%
R version (sandbox)	R version 4.5.3 (2026-03-11)
Platform (sandbox)	x86_64-pc-linux-gnu
Sandbox image	rval-sandbox:0.1.0
CRAN snapshot	2026-04-15
Duration	36.32 s
Finished at (UTC)	2026-09-05T19:52:13Z

IQ — environment and spec fingerprint

R version (sandbox)	R version 4.5.3 (2026-03-11)
Platform (sandbox)	x86_64-pc-linux-gnu
Sandbox image	rval-sandbox:0.1.0
Spec sha256	c8b0888d9f9a6af7b9b3fbb3db6b1b21430b4b8e3ffb6b3233180adecece435f
Spec config_version	0.1.2

OQ — requirement execution

Validation status: **pass** — cases pass 5, diff 0, fail 0, error 0.

Requirement	Risk	Linked cases	Status
REQ-001: toJSON serializes a known R structure to the exact RFC 8259 JSON text (known-answer acceptance criterion).	high	G, M, -, 0, 0, 1	pass
REQ-002: fromJSON(toJSON(x)) is the identity for numeric vectors and data frames (metamorphic roundtrip acceptance criterion).	high	MR-001, MR-002	pass
REQ-003: Generated numeric vectors survive the JSON roundtrip within tolerance for arbitrary contents and lengths (property-ba...	medium	P, B, -, 0, 0, 1	pass
REQ-004: Malformed JSON input raises an error, never crashes the process and never returns silently wrong data.	medium	F, Z, -, 0, 0, 1	pass

PQ — intended-use scenarios

Scenario	Description	Status
PQ-001	Analysis-pipeline payloads round-trip through JSON without value drift.	pass

Requirement traceability matrix

REQ	Case	Type	Result	Detail
REQ-001	G	—	untested	
REQ-001	M	—	untested	
REQ-001	-	—	untested	
REQ-001	0	—	untested	
REQ-001	0	—	untested	
REQ-001	1	—	untested	
REQ-002	MR-001	metamorphic	pass	50/50 cases held (relation=roundtrip_identity, generator=gen_numeric)
REQ-002	MR-002	metamorphic	pass	30/30 cases held (relation=roundtrip_identity, generator=gen_dataframe)
REQ-003	P	—	untested	
REQ-003	B	—	untested	
REQ-003	-	—	untested	
REQ-003	0	—	untested	
REQ-003	0	—	untested	
REQ-003	1	—	untested	
REQ-004	F	—	untested	

REQ	Case	Type	Result	Detail
REQ-004	Z	—	untested	
REQ-004	-	—	untested	
REQ-004	0	—	untested	
REQ-004	0	—	untested	
REQ-004	1	—	untested	
—	GM-001	golden	pass	identical
—	PB-001	property	pass	predicate held for 50/50 seeded cases
—	FZ-001	fuzz	pass	malformed: pass (graceful error); wrong_type: pass (graceful error); empty_string: pass (graceful error)

Mutation testing (suite defect-detection power)

Mutants 20 (killed 4, survived 16, timeout 0)
Mutation score **0.20** Surviving mu-
Operators arith_swap, relop_swap
 tants (behavior changes this suite failed to detect):

Mutant	File	Line	Operator
M-001	R/collapse.R	25	arith_swap
M-002	R/pack.R	117	relop_swap
M-004	R/simplifyDataFrame.R	38	relop_swap
M-006	R/unescape_unicode.R	8	relop_swap
M-009	R/simplify.R	126	arith_swap
M-010	R/asJSON.raw.R	17	relop_swap
M-011	R/asJSON.complex.R	17	relop_swap
M-012	R/asJSON.factor.R	6	relop_swap
M-013	R/apply_by_pages.R	9	arith_swap
M-014	R/asJSON.scalar.R	3	relop_swap
M-015	R/asJSON.function.R	5	relop_swap
M-016	R/asJSON.raw.R	14	relop_swap
M-017	R/apply_by_pages.R	10	arith_swap
M-018	R/asJSON.POSIXt.R	14	relop_swap
M-019	R/simplifyDataFrame.R	68	relop_swap
M-020	R/fromJSON.R	87	relop_swap

automated validation verdicts are decision-support evidence and never replace qualified human review and sign-off; the mutation operator set is finite, so the mutation score is an upper-biased estimate of true defect-detection power

Annex: System Information

Generator: rval-gateway 0.7.0 (Python stdlib); R version 4.5.3 (2026-03-11); {riskmetric} 0.2.7; decision config v1.1.0 (sha256 531ab81389ab28ca59dc74ef)
 evidence collection mode: network.

Evidence collected at (UTC) 2026-09-05T19:47:14Z
Assessment scored at (UTC) 2026-09-05T19:47:13Z
Network sources ok / failed 3 / 0

Approval Timeline

Audit-chain events for this package (assessment → electronic signature → repository publish), in chain order; details carry the sha256 record-hash prefix of each tamper-evident audit record.

Event	Actor	Timestamp (UTC)	Detail / record
assessment	dev-mode	2026-09-05T19:47:14Z	GO-WITH-CONDITIONS (tier gxp-support, score 0.2829) [663a6897511c]
tier2	dev-mode	2026-09-05T19:48:13Z	tier2 check-errors — tests 63/68, coverage 65.2%, check errors — validation spec present, approval pending — smoke mode; validation section suppressed [404a7330d65b]
esign	qa.reviewer	2026-09-05T19:51:34Z	signed 'approved' for jsonlite spec v0.1.2 [1ae4fe5e71ce]
tier2_validation	qa.reviewer	2026-09-05T19:53:19Z	tier2 validation pass (req 4/4, mutation 0.2) [835495ec3032]

Provenance. Generator: rval-gateway 0.7.0 (Python stdlib). Scores computed by an R subprocess: R version 4.5.3 (2026-03-11), {riskmetric} 0.2.7 (two-process pattern; code-executing metrics excluded — assessed package code was not executed). Decision config v1.1.0 (sha256 531ab81389ab). Evidence collection mode: network. Hard rule: no statement without an evidence ID; sections lacking evidence are suppressed. Tier-1 screening output under a two-tier strategy; metrics inform a qualified human's assessment and are not a proof of correctness.